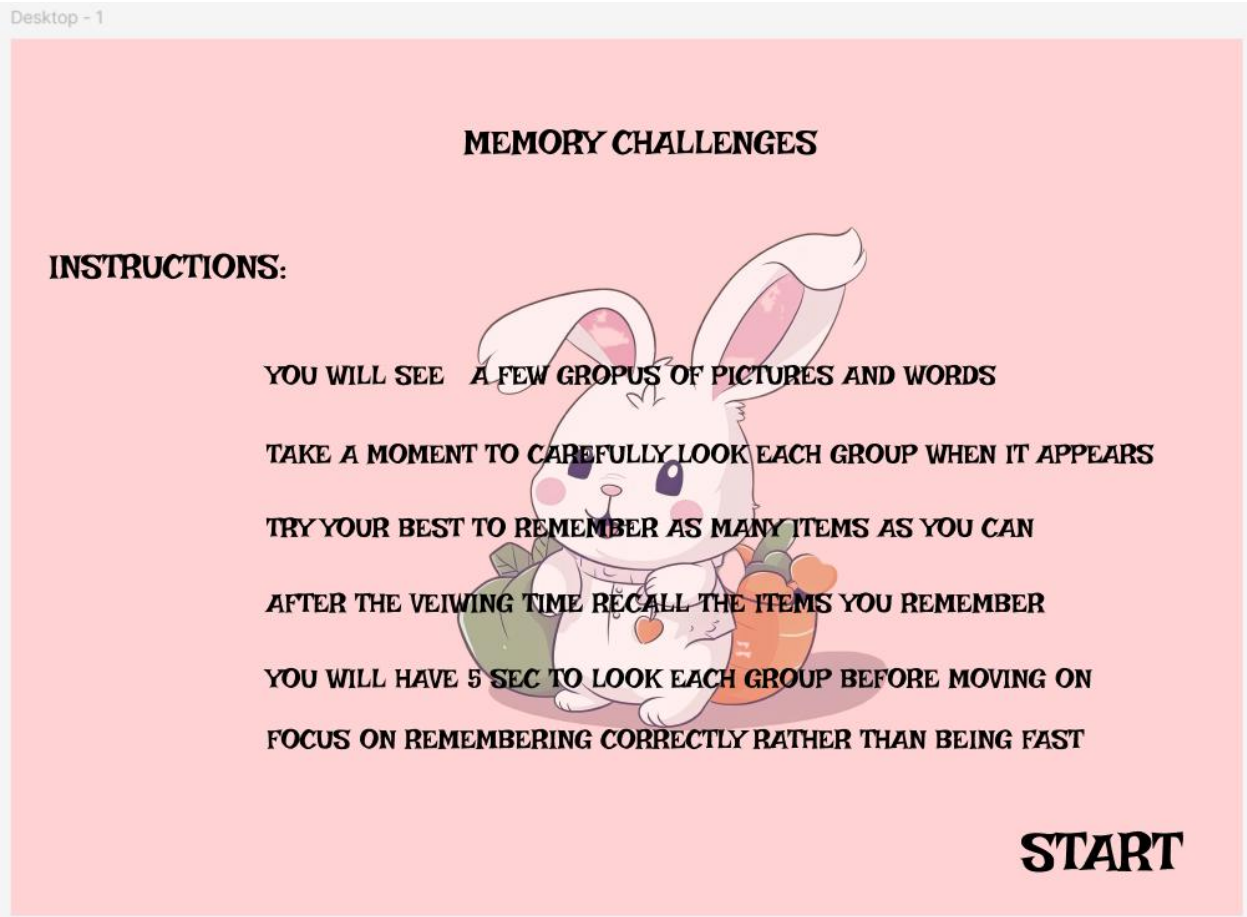
USER INTERFACE AND DESIGN EXP-2:-

Introduction: Chunking is a cognitive strategy that involves breaking information into smaller, meaningful units. This experiment uses a game-based UI to analyze how chunking improves short-term memory recall by presenting visual elements within a limited time.

FRAME 1:

INSTRUCTION PAGE



FRAME 2:-

CHUNKING PHASE Analysis of Chunking Phase Screen

Purpose: This is the memory encoding stage where users observe icons and mentally group them. UI Elements:

• Countdown Timer (10 seconds) to create urgency.

• Progress bar to visualize remaining time.

• Grid of icons arranged to support grouping by category, color, or similarity. Working: Users quickly scan and form mental chunks before time ends.

Cognitive Benefit: Chunking reduces memory load and improves recall efficiency.

Desktop - 2

4

MEMORY CHALLENGES

REMEMBER THESE CARTOONS

3

MEMORY CHALLENGES

REMEMBER THESE CARTOONS

Desktop - 4

2

MEMORY CHALLENGES

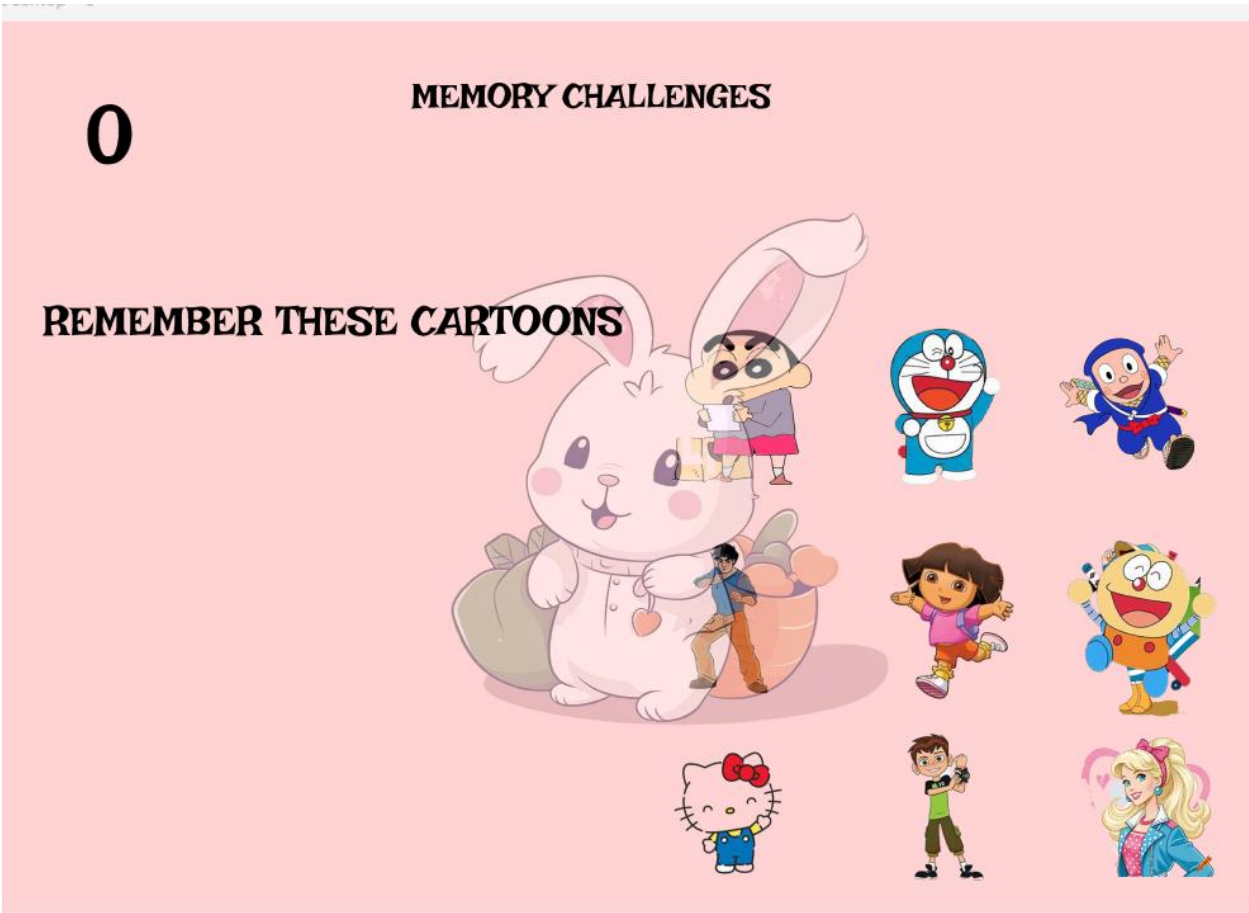
REMEMBER THESE CARTOONS

Desktop - 5

1

MEMORY CHALLENGES

REMEMBER THESE CARTOONS



FRAME 3:-

RECALL PHASE Analysis of the Recall Phase

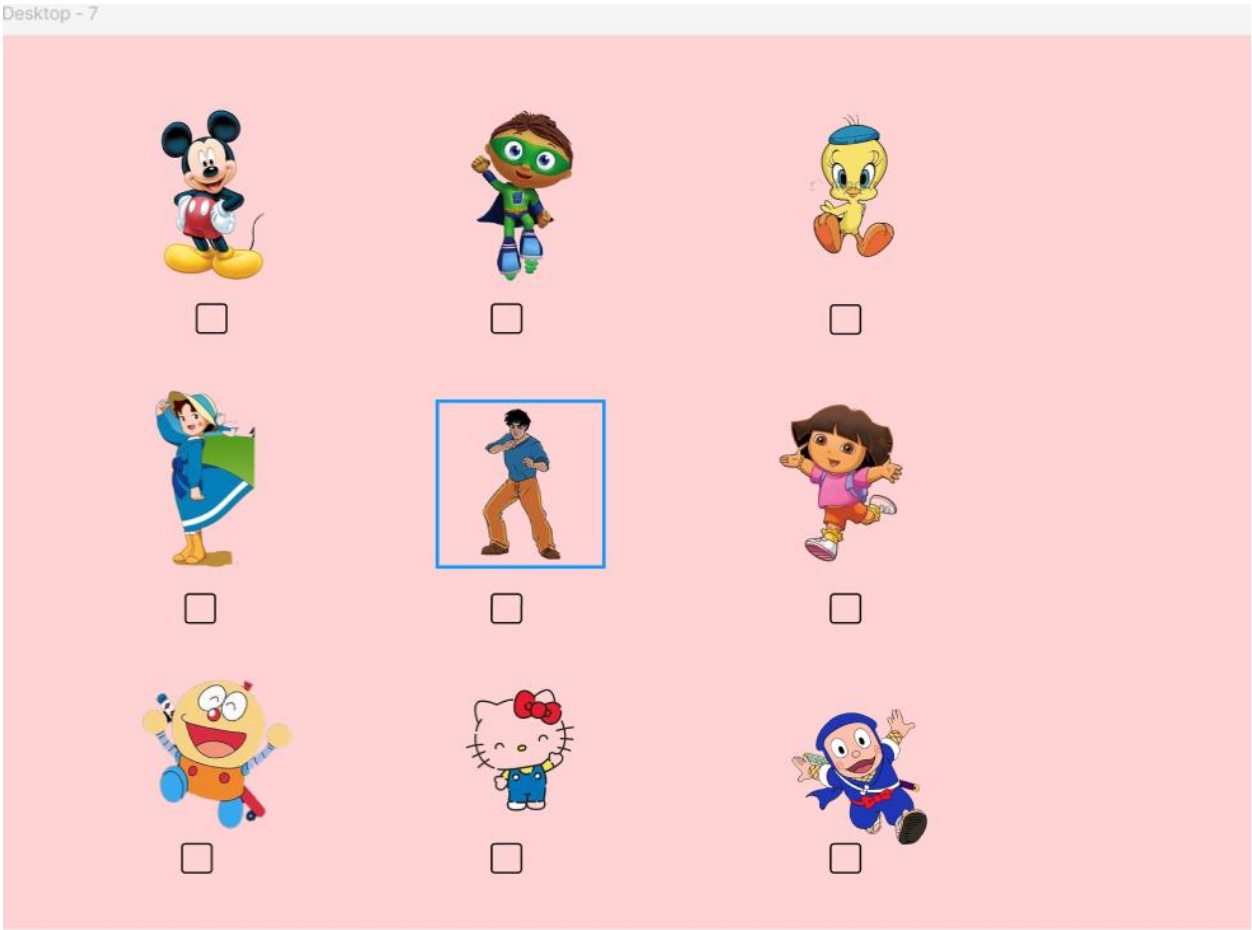
1. Purpose • Tests memory by asking users to recall previously seen icons. • Measures the effectiveness of chunking.

2. UI Elements • Title & Instructions: Clearly explain the task. • Multiple Choice Icon Grid: o Contains both correct icons and distractors. o Users must identify previously seen items.

• Radio Buttons: o Easy and clear selection method. • Submit Button: o Confirms user choices.

3. Working Principle 1. Users recall icons from memory. 2. Select remembered icons. 3. Distractors test memory accuracy.

4. Submit leads to result page.



FRAME 4:-

RESULT PAGE Analysis of the Result Page

1. Purpose • Displays recall accuracy. • Provides feedback and next actions.

2. UI Elements • Score Display (e.g., 6/9): o Shows performance clearly. • Action Buttons: o Play again: Move to next level. o Home: End the activity.

3. Working Principle 1. System evaluates selected icons. 2. Calculates score. 3. User chooses next action.

4. Cognitive & UX Benefits • Immediate feedback improves learning. • User control enhances experience. • Gamification reduces stress and increases motivation.

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